Lecture 23 · Part VI · Lecture notes

Vision transformers and multimodal retrieval

Dataset: COCO, 200 catalogue entries

How to use these notes

The primary text for this lecture is Géron, Chapters 15–16. These notes are not a replacement for it and do not repeat it: they are the lecture's own argument written out at length — the derivation done slowly, the numbers with the runs that produced them, and the reasoning between one slide and the next that a deck can only gesture at. Read the chapter for the method; read these for why the lecture makes the choices it does. Every figure quoted here is printed by this lecture's notebook, and the course checks that mechanically rather than trusting it.

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1 The first brief whose input and output differ in kind

Part V built retrieval over text and over user behaviour, and both reduced to the same object: two encoders and an inner product. Today the two towers take *different modalities* — one encodes images, the other text, and a photograph and its caption must land close together.

A retailer's catalogue team: the entries are photographs. Product photos arrive with an identifier, a price and a picture; written descriptions are inconsistent, often absent, and never written by the person searching. *Customers type sentences. The catalogue stores pixels.* Nothing in the system converts one into the other, and "label every image" does not either: somebody has to write the labels, and the customer will use a word that is not in the label set.

Input	Output
a sentence typed by a customer "a red bus on a city street"	a ranked list of catalogue entries the entries whose picture shows that

No labels, because nobody is going to annotate the catalogue; no fixed vocabulary, because the query is whatever the customer typed; and the answer must be a ranking, not a yes or no. This is the first brief in the course whose input and output are not the same kind of object.

Keyword search needs text on both sides, and so does the semantic search of the previous lecture — it embeds a sentence and compares it to *other sentences*. An entry with no description is not badly ranked by a text index. It is *absent*. Hold that thought: the next lecture counts how many entries it costs, and repairs it.

1.1 The corpus, and the second corpus

200 catalogue entries, 32.6 MB, taken in image-id order from COCO's 2014 validation set so that everyone's catalogue is the same catalogue, each with five independent human captions written by people who were not shown each other's. Every retrieval number in this lecture is measured against 200 candidates.

Failure condition — the candidate set is the number

Every retrieval score here is measured against 200 candidates. Widen the catalogue and the same model reports a worse number without having changed, so a recall figure quoted without its candidate-set size cannot be compared with anything. Two hundred is small, and small is fine; pretending it is larger is not.

Two different uses for five captions

Caption 1 is the entry's *description*, when it has one — what a text index would see. Caption 2 is the *customer's query*. Captions 3 to 5 are held back.
Using two different people's sentences about the same picture makes this a paraphrase problem rather than a lookup. If we used the same sentence on both sides, a hash table would score 100%.

The catalogue has no labels, so it cannot play the role of a known-answer test. A second corpus does: 500 CIFAR-10 test images and ten class names — ten labels we actually have. CIFAR-10 is 32×32, which is a hard setting for any model trained on web photographs, and we will see the cost.

1.2 An image as a sequence

A transformer takes a sequence and an image is a grid, so cut it up: split into fixed patches, flatten each and project it linearly — *that projection plays the part of a language model's token-embedding table, with patches in place of tokens* — add a positional encoding, because attention by itself has no notion of order and a grid has two, then run the standard blocks. A 224×224 image at 16×16 patches is $14 \times 14 = 196$ tokens, well within reach of the quadratic cost, which is exactly why that patch size and not a smaller one.

2 A metric, and a baseline that needs no experiment

The system returns a ranked list, so what matters is where the right entry landed. For query q with relevant entry $q^\star$,

$$r_q = 1 + \#\{\, j \neq q^\star : s(q,j) \geq s(q,q^\star) \,\}, \qquad \mathrm{R@}k = \frac{1}{Q}\sum_{q=1}^{Q} \mathbf{1}[\, r_q \leq k \,].$$

Ties count *against* us — that is the $\geq$ — because a metric that flatters a degenerate score matrix is not a metric. With exactly one relevant entry per query, recall and precision at k carry the same information, and recall is the one the literature reports. We read R@1, R@5 and R@10 — R@1 being the number the product cares about, because it is the top of the page.

Beside recall goes the mean reciprocal rank:

$$\mathrm{MRR} = \frac{1}{Q}\sum_{q=1}^{Q} \frac{1}{r_q}.$$

Rank 1 contributes 1, rank 2 contributes 0.5, rank 100 contributes 0.01: it rewards being nearly right and stops rewarding it quickly, which matches a page of ten results.

Every number needs a baseline, and here the baseline needs no experiment.

The trivial baseline, computed rather than run

Shuffle the catalogue: one relevant entry among n, placed uniformly at random:

$$\Pr[r \le k] = \frac{k}{n}, \qquad \mathbb{E}[r] = \frac{n+1}{2}, \qquad \mathbb{E}\left[\tfrac{1}{r}\right] = \frac{H_n}{n}.$$

No simulation, no seed, no error bar — this one is arithmetic.

At $n = 200$ that is R@1 of 0.5%, R@5 of 2.5%, R@10 of 5.0%, an expected rank of 100.5 and an MRR of 0.029.

3 Two encoders, trained apart

Encode the images with a ViT trained on ImageNet labels — no text anywhere in its training — taking the `[CLS]` vector as the image. Encode the queries with the sentence encoder from the previous application, mean-pooled over the real tokens only (forget the attention mask and you average in the padding).

One line of the image encoder is worth a note: the checkpoint has no pooler weights, so it is loaded with `add_pooling_layer=False`. Ask for a pooler and you get a randomly initialised layer, silently.

Then take the cosine — except that the image vectors live in 768 dimensions and the sentence vectors in 384, so `V @ T.T` does not even run: there is no inner product between the two spaces. Make the dimensions agree three ways, so nobody can blame the fix: a random Gaussian projection from 768 to 384, keeping the first 384 image coordinates, or zero-padding the text vectors to 768.

A similarity is meaningless alone, so compute two numbers: the cosine between an image and its own caption, and between an image and somebody else's — nearly forty thousand of the latter.

System	R@1	R@5	R@10	Median rank
Random ranking (exact)	0.5%	2.5%	5.0%	100
ViT + MiniLM, random projection	1.0%	4.0%	6.0%	97
ViT + MiniLM, truncated	0.5%	2.0%	5.5%	88
ViT + MiniLM, zero-padded	0.0%	2.5%	5.0%	88

Two of the 200 queries put the right image first. Shuffling is expected to get one. Two strong encoders, over a hundred million parameters between them, and that is the whole difference.

Say the result out loud

A picture of a bus and the sentence "a red bus on a city street" are no closer than that picture and a sentence about a sandwich.

Both encoders work perfectly. The ViT separates images from one another; MiniLM separates sentences from one another. Neither was ever shown a single (image, sentence) pair, so neither has any reason to place them anywhere in particular relative to each other.

Why, precisely: an embedding space is defined only up to whatever the training objective constrained. ImageNet classification constrains images relative to images; sentence-similarity training constrains sentences relative to sentences; and *nothing in either loss ever compared a vector from one to a vector from the other*. Apply any orthogonal map to the image space and every image–image cosine is unchanged, so the ViT's loss is unchanged — while every image–text cosine changes. The loss cannot distinguish those worlds, so it did not pick one. The fix is not a better image encoder; it is a training signal that ever compared the two.

4 Derivation: the sphere, and the temperature

A jointly trained pair of encoders scores a batch of B pairs, image tower f and text tower g, each ending in a linear map into $\mathbb{R}^d$:

$$\mathbf{u}_i = \frac{f(x_i)}{\|f(x_i)\|}, \qquad \mathbf{v}_j = \frac{g(t_j)}{\|g(t_j)\|}, \qquad S_{ij} = \mathbf{u}_i^\top \mathbf{v}_j,$$

a $B \times B$ matrix of cosines. S_{ii} is a pair that belongs together; every off-diagonal entry is a pair that does not. *The loss only ever sees this matrix.*

4.1 Why normalise

The unnormalised inner product factorises as $\|a\|\|b\| \cos\theta$, mixing two quantities: which direction the encoder chose, and how loudly it said it. Only the direction carries the semantics — the length is whatever the last linear layer happened to scale to, and on our catalogue the longest image embedding is 1.32 times the shortest. Divide both by their norms and S_{ij} is a pure cosine in $[-1, 1]$, comparable across pairs, models and runs.

4.2 Two modalities, two regions

Measured on the jointly trained encoder used later in this lecture, images occupy one region of the sphere and captions another, with the two centroids 0.83 apart. The objective aligns pairs *relatively* and never asks the two clouds to coincide, so nothing in training closes the modality gap. One consequence: an absolute cosine threshold tuned on image–image pairs means nothing for image–text pairs.

4.3 What the temperature does

Row i of S is a B-class softmax whose correct answer is column i, with every cosine divided by a temperature τ:

$$\mathcal{L} = -\frac{1}{B} \sum_i \log \frac{\exp(S_{ii}/\tau)}{\sum_j \exp(S_{ij}/\tau)}.$$

This is the InfoNCE loss.

Differentiating gives the softmax's familiar prediction minus target, scaled by $1/\tau$: $\partial \mathcal{L}_i / \partial S_{ij} = \frac{1}{\tau}(p_{ij} - \mathbf{1}[j = i])$. So τ does not change which column is largest — *it cannot change the top-1*

accuracy of a fixed model — it changes where the gradient goes. Small τ concentrates the weight on the few hardest negatives.

Too small a τ, and the loss is dominated by one negative per row, which may be a mislabelled pair rather than an informative one.

5 One space, two modalities

The idea in one paragraph: take a large collection of (image, caption) pairs. Push each image through an image tower and each caption through a text tower, both ending in a linear map into the same d-dimensional space. Normalise onto the unit sphere. Then train so that, within a batch, each image's own caption scores higher than every other caption in the batch. The derivation above said why each of those choices; here we use the result and measure it.

A jointly trained dual encoder — a ViT-B/32 image tower, a twelve-layer text tower, and a shared 512-dimensional space on the unit sphere, trained on image–text pairs collected from the web. The whole system is fourteen lines, and the only difference from the separately trained pair is that the two sides now have the same shape, so the matrix product exists.

5.1 The known-answer test first

The catalogue has no labels, so nothing there can tell us the model is loaded correctly, pointed at the right tensors, and preprocessing images the way it was trained to. So: 500 CIFAR-10 images, ten class names, embed "a photo of a {class}" for each, and assign each image to the nearest sentence. *The classifier is ten sentences, and there is no fitted parameter anywhere.* No training, no fitting, no labels used except to score: that is what "zero-shot" means, and here it is the check, not the product.

90.6% over 500 images against a 10.0% chance level, from a model that has never seen CIFAR-10.

The sentence is a hyperparameter

Prompt	Accuracy over 500 images
"dog"	88.0%
"a photo of a dog"	90.6%
"a low-resolution photo of a dog"	90.6%

A spread of 2.6% from changing the wording alone. *A "zero-shot" number is a number for one particular sentence, and the sentence is a choice you made.*

5.2 Back to the catalogue

Same 200 images, same 200 queries, same metric, same ties rule, same baseline — only the encoders changed.

System	R@1	R@5	R@10	Median rank	MRR
Random ranking	0.5%	2.5%	5.0%	100	0.029
ViT + MiniLM	1.0%	4.0%	6.0%	97	0.037
Joint dual encoder	75.0%	97.0%	99.0%	1	0.841

Text to image, 200 candidates, ties against the model, one caption per entry as the query. *Two of those three rows are the same row.*

5.3 Report the failure cases

75.0% at rank 1 means 25.0% of queries did not get the right entry first, and they fall into groups: queries describing a scene rather than an object, where several entries fit equally well and the "wrong" one is often a reasonable answer; fine distinctions the caption makes and the picture barely shows; and counts and spatial relations — "three", "behind", "to the left of".

One relevant entry is a fiction

Our metric assumes exactly one right answer per query because that is what the data gives us. For "a man riding a horse" on a real catalogue it is simply false, so the measured recall is a *lower bound*.

And customers do not type captions. They type "something to sit on", "gear for bad weather", "something you would take to the beach". The right output there is not one entry but a shortlist with reasons — and we have no way to produce, or to score, a reason. That is the second thing the next lecture repairs.

5.4 Six ways to leak in a retrieval evaluation

The query text is also the indexed description, so you are measuring string equality. The query's own entry is scored against a shortlist built using the query. The prompt template was chosen by looking at the number it produced. Recall is reported without the candidate-set size. Ties are broken in the model's favour. Or normalisation is applied to one side only, so "cosine" is not a cosine.

Failure condition — the tie bug, which reports 100%

Ranking with a strict inequality counts only candidates that *beat* the right answer, so anything tying with it is placed behind it for free.

Feed that a similarity matrix of all zeros — a model that has learned nothing, or a bug that zeroed the embeddings — and every rank is 1. It reports R@1 = 100%.

And the third leak above is a hyperparameter chosen on the test set, with a prompt template in place of a numeric hyperparameter: three templates tried, the best on the test set reported. The spread across the three was 2.6%, so the optimism is of that order, and the repair is the one you already know: choose the template on a validation split, and report the chosen one on the test set, once.

6 The notebook

What to change

Skip the normalisation before the inner product. Retrieval degrades, and the reason is in the derivation: without unit norm the score confounds direction with magnitude, and magnitude carries no meaning here.

7 Design exercise: a stock-photo company's text-to-image search

The lecture closes with a design exercise in the manner of an ML system-design interview, on a product with the shape of data this model suits.

The task. A stock-photo company licenses photographs to designers and marketers, and its search page is where every licence starts. A designer types "two colleagues laughing over coffee" and the page must return a ranked page of photographs whose pixels show that. Contributors upload each photo with a caption and keywords written to be found rather than to be accurate; the designer's words are never the contributor's. The business measures downloads per search, searches that end without one, and licence revenue. Given: the library — photographs, illustrations and vector art, each with its contributor's caption and keywords — the editors' category labels on a subset, and the search log with its downloads. Tens of millions of images, uploads every minute, a page of twenty results per query. The output feeds the search page and the editors who curate the front page.

1. **Frame it.** The output is a ranking, so the quantity is where the relevant picture landed: recall at a cut-off, with the cut-off at the page the designer sees, and the mean reciprocal rank beside it. Ties count against the model, so that a constant scorer reports what shuffling reports, and the baseline is arithmetic rather than an experiment: for one relevant picture among n, recall at k under random ranking is k/n. *Pitfall:* the candidate set is the number. A recall quoted without the size of the set it was measured against cannot be compared with anything: three in four at rank one over a few hundred candidates is not three in four over the library, the same model reports a worse number as the haystack grows without having changed, and a launch promised on the small number disappoints. Say the size every time you say a recall, and measure at the library's size.

2. **The pairs.** The training signal is a pair, a picture and a sentence that describes it, and the library has millions of them for free in the contributors' captions. But a contributor's caption is written to be found: a list of keywords, the brand of the camera, a mood the picture does not show, the same sentence on every frame of a shoot. For the judged queries the editors write what a designer would type, and the search log supplies real queries with the downloads that followed them as judgements. *Pitfall:* measuring with the contributor's own caption as the query. The text index already holds that string, so the system is scored on string equality and a hash table would be perfect. Use a different person's sentence about the same picture on the query side, so that the task is paraphrase rather than lookup, and clean the keyword-stuffed captions before pairing, because the model learns what the pairs say.

3. **Split by shoot.** Hold out the judged queries and their pictures as the test set, and carve a

validation set from the rest for every choice — prompt wording, temperature, cut-off. A library is organised in shoots, dozens of near-identical frames of the same scene with the same caption and contributor: split by shoot, so that no frame of a test scene is in the training pairs, and read the test set once. *Pitfall:* splitting by picture. Another frame of the same shoot sits in the training pairs, and the model has seen the test answer up to a change of angle — a leak of the standing kind, with no column duplicated. The recall is splendid, it measures memory of the shoot, and the review approves an untested model. Group by shoot and contributor, and split the groups.

4. **One sphere.** The image tower is a vision transformer — the picture cut into patches, each projected like a token, a positional encoding, the standard blocks — and the text tower a transformer over the words. Both end in a linear map into the same space and both outputs are normalised, so the score is a pure cosine: direction carries the meaning, length is whatever the last layer scaled to. Every picture's vector is precomputed once; at query time the sentence is encoded and the inner products taken. *Pitfall:* two strong encoders trained apart — a vision transformer from image labels, a sentence encoder from sentence pairs — glued together by matching their dimensions. Nothing in either loss ever compared a vector from one with a vector from the other, so their relative orientation is arbitrary and the retrieval is indistinguishable from shuffling. The fix is not a bigger encoder; it is a training signal that saw both at once.

5. **The batch.** A batch of B pairs gives a $B \times B$ matrix of cosines; row i is a B-class softmax whose right answer is column i, and every other caption in the batch is a negative for free. The cosines are divided by a temperature τ before the softmax, which cannot change which column is largest but decides where the gradient goes — small τ sends almost all of it to the few hardest negatives — and τ is chosen on the validation split with the batch composition held fixed. *Pitfall:* a false negative in every batch. Batches drawn in upload order hold several frames of the same shoot with the same caption, so the objective pushes a picture away from a sentence that describes it, and with a small τ that one mislabelled pair takes the whole row's gradient. Draw batches across shoots, one frame per shoot, and keep τ away from the floor where a single negative dominates.

6. **A known-answer test.** The library has no labels that say "this is the right picture", but a subset has the editors' category labels, and that is the known answer: embed "a photo of a {category}" for each and assign every labelled picture to the nearest sentence. No fitted parameter anywhere; it is the check that the model is loaded, pointed at the right tensors and preprocessing pictures as it was trained to, read per category and per content type. *Pitfall:* the sentence is a hyperparameter. Trying several wordings and reporting the best on the test labels is a hyperparameter chosen on the test set, with a prompt in place of a number, and the spread between wordings is the size of the optimism. Choose the wording on the validation split and report it on the test once; and a checkpoint that silently loads a randomly initialised head still prints a number, which the known-answer test catches before a customer does.

7. **Domain shift.** Same metric, same ties rule, same random baseline, on the editors' judged queries over a candidate set the size of the library's: recall at the page and the reciprocal rank, read once. The model was pretrained on web photographs and captions, and the library is studio shots on white, illustrations, vector art and queries about mood that no web caption contains; the failure cases — scenes with several reasonable answers, fine distinctions, counts and spatial relations — are reported. *Pitfall:* one mean over a web-like benchmark, carried

to the library. Slice the recall by content type and by query type: a model that puts the right photograph first three times in four can rank illustrations no better than chance, the mean hides it, and illustration downloads fall. And one relevant picture per query is a fiction on a real library, so the measured recall is a lower bound — judge several, and say so.

8. **In production.** The index is the matrix of picture vectors and the model version that produced them. A new upload is one encoder pass; a new model re-encodes the library, built beside the old index and switched once, behind an A/B test read on downloads per search. The known-answer test and the judged queries are re-run after every rebuild, a drop being the regression owned by whoever is on call, and a fixed set of queries about people — "a doctor", "a family" — is audited for who the first page shows. *Pitfall:* who the model retrieves. The pairs the model learned from came from the web, whose captions attach some people to some words; recall gives full credit for a first page that answers "a doctor" with one kind of person every time, nothing in the objective notices, and the customer who does is a journalist. Measure it as a distribution over the first page for the audit queries, make it a requirement the editors sign off, and re-check it on every refresh.

9. **Ship it.** Uploads with captions → captions cleaned, editors' queries on the query side → split by shoot → two towers into one space, both normalised → contrastive training, batches across shoots → known-answer test → recall at the page on the judged queries, sliced by content type → A/B test, index with its model version, audit queries re-run.

8 Exercises

The five set at the end of this lecture's deck, in the style of the written paper. The solutions are on Lecture 24's deck and in the exercise book.

1. Two encoders are trained separately, one on images and one on text. Explain why their cosine similarity carries no information about matching, using a rotation argument. [5]

2. A rank is computed by counting only the candidates that score strictly higher than the right answer. State the Recall@1 this reports for a similarity matrix of all zeros, and the repair. [5]

3. The InfoNCE logits are cosines divided by a temperature τ. State what τ cannot change for a fixed model, and what it does change. [5]

4. Recall@10 is reported over 200 candidates. State the exact random baseline, and why it is computed rather than simulated. [3]

5. An entry with no written description is never returned by a text index built over the descriptions. State whether this is a ranking failure or a structural one, and what follows. [4]

Summary

An image becomes a sequence of patches: cut into 16×16 pieces, each projected like a token, with a positional encoding because attention has no notion of order. Retrieval is measured by

where the right entry lands — Recall@k and the mean reciprocal rank, ties counted against the model — beside a random baseline that is arithmetic rather than an experiment.

Two separately trained encoders produce spaces that are *unrelated*, and the retrieval built from them is indistinguishable from shuffling. Not because either encoder is bad, but because no loss ever compared a vector from one with a vector from the other, and an orthogonal map on the image space leaves that loss unchanged while changing every image–text cosine.

Normalise, because only direction is semantic. Even trained jointly, images and captions keep separate regions of the sphere, so an absolute threshold does not carry from one kind of pair to the other. Divide by a temperature τ: it cannot change a fixed model's ranking, only which negatives the gradient hears.

Jointly trained, the same measurement gives R@1 of 75.0% against 1.0% over 200 candidates, and the known-answer test reaches 90.6% zero-shot on CIFAR-10 — for one particular prompt, out of three that span 2.6%. Then audit the evaluation: the failure cases, the six leaks, the tie rule.

The closing design exercise built a stock-photo company's text-to-image search in nine steps — recall with its candidate-set size, another person's sentence as the query, a split by shoot, two towers on one sphere, batches drawn across shoots with τ off the floor, a known-answer test with its wording chosen on validation, recall sliced by content type, and a release behind an A/B test with the index versioned and an audit of who is retrieved — each step named for the pitfall it avoids.